CS425 · CONVERSATIONAL SYSTEMS

Jargonaut

Evidence-Grounded Contract Simplification



[TODO] Team G1T5 · names · date

Contracts create a comprehension gap

People sign contracts without fully reading or understanding them.

LEGAL LANGUAGE

Long documents with unfamiliar legal terminology

USER NEED

Find the relevant evidence and explain it clearly.

Build clarity without losing the evidence

Upload a contract. Ask a question. Get a plain-English answer grounded in the source.

SCOPE
English commercial contracts

TOPICS
10 CUAD categories

RESEARCH QUESTION
Does extraction plus simplification improve understandability and keep meaning?

Three datasets, three jobs

01

CUAD PDFs

Evidence extraction and end-to-end evaluation

02

20 labelled clauses

Controlled validation of plain-English rewriting

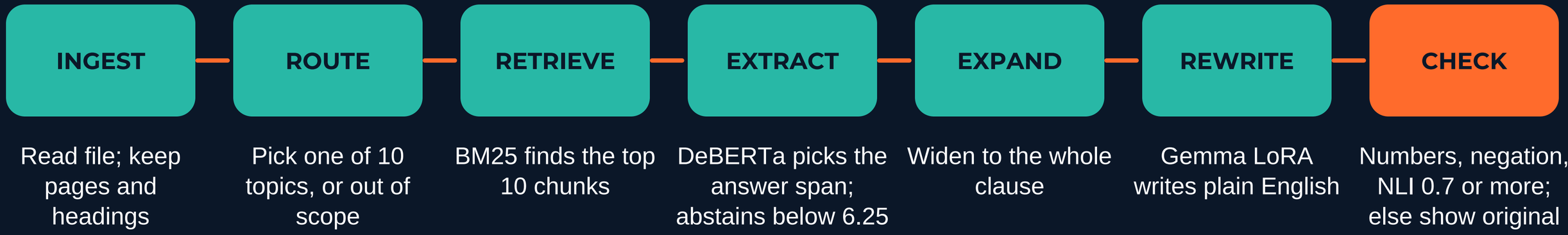
03

10 fictional contracts

Labelled before any model ran; for demonstration

[TODO] Confirm dataset name (SimpLegal-Pref or SimpleGAL) and whether ContractNLI was used

Seven steps keep every answer traceable



Evidence is preserved through retrieval, extraction and rewriting — then checked before display.

Models and key settings

Component	Model / method	Key setting
Router	Gemma 4 E4B, adapter off	Asks if runner-up is 25% or more
Retriever	BM25	top-k: 10
Extractor	DeBERTa-v3-large, tuned on CUAD	threshold: 6.25
Rewriter	Gemma 4 E4B + LoRA (Run 6)	rank 16, lr 2e-4, 886 pairs
Checker	NLI model + number/negation rules	NLI coverage 0.7 or more

[TODO] DeBERTa epochs, batch size, learning rate; why E4B instead of 31B

Beyond running open-source code on CUAD

01

Evidence-first architecture

Answers retain a visible path back to source text.

02

Calibrated UX

Model confidence becomes a probability users can interpret.

03

End-to-end evaluation

Ingest to check run on CUAD test PDFs; routing tested separately.

Evidence extraction on 102 test contracts

102

test contracts

83.5%

overall recall

97.4%

precision

CUAD test PDFs: 1,020 questions, 443 with evidence. Span hit 81.9%; BM25 top 10 holds the gold evidence for 90.1%.

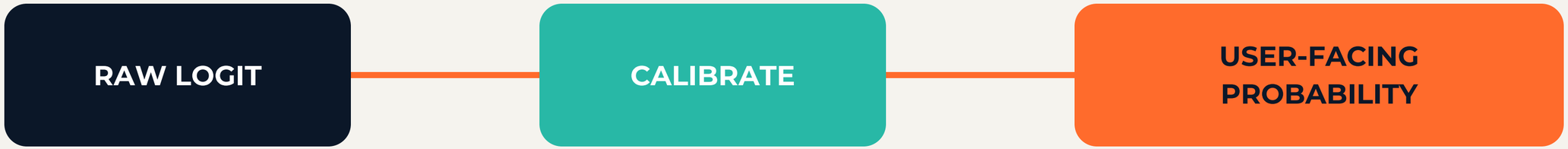
Recall varies by clause category



Recall on CUAD test PDFs. Orange: four weakest categories, each with fewer than 35 test questions with evidence.

From a raw score to readable confidence

$$p = 1 / (1 + e^{-(0.326 \times score - 1.012)})$$



Platt scaling fitted on validation; 0.74 at the 6.25 threshold. Test: ECE 0.039, Brier 0.045. Only 0.8 or more is labelled High.

Question routing: two approaches on test

Router	Accuracy	Latency	Failure mode
Gemma (the app's)	97.3%	about 0.2 s	3 errors in 110 questions
Keywords + NLI	76.4%	[TODO]	22 in-scope wrongly rejected

110 test questions written blind from category descriptions; router run on test once.

Plain-English rewriting · validation on 20 clauses

ORIGINAL CLAUSE

FKGL grade 28.75
SARI 37.19
NLI coverage 0.95

ZERO-SHOT GEMMA

FKGL grade 7.96
SARI 46.95
NLI coverage 0.73
Number precision 0.88
Negation kept 0.80

RUN 6

FKGL grade 8.74
SARI 41.49
NLI coverage 0.79
Number precision 1.00
Negation kept 1.00

[TODO] Run 6 seeds 1 and 2, test-set result, and the manual review sheet (20 clauses) are still to do

Rewrites in the full system

380

accepted clauses

Evidence extracted, expanded, rewritten and checked end to end on CUAD test PDFs.

53.7% pass all three checks and are shown in plain English
Reading grade drops from 18.9 to 8.3
Numbers kept 93.7%, negation kept 94.2%
NLI coverage is the only failing check for 138 of the 176 held back

The checker caught a real error



BEFORE · MODEL CLAIM

Clause: "either party may terminate this Agreement at any time upon 30 days written notice"
Rewrite: "Neither party can end this Agreement at any time if no default exists"

AFTER · CHECKER RESPONSE

NLI coverage was 0.00, so the app showed the original clause with a warning instead of the rewrite.

A fictional test set designed before running

10
invented contracts

10
topics, one per contract

100
contract × topic pairs

All 100 pairs were labelled before any model ran.

[TODO] Add performance on the 100 labelled pairs

Why did this happen?

WHAT WORKED

High precision: 97.4% of accepted answers are real evidence.

The checks caught a reversed termination right.

WHAT DID NOT

Plain-style wording: 8 of 10 Governing Law clauses missed.

Short clauses often fail the 0.7 NLI check: 44% pass.

Limitations become the roadmap

01

Sample size

Weak categories have fewer than 35 test questions with evidence

02

Run 6

Validation results only; seeds 1 and 2 and test set pending

03

Calibration

Probabilities from 0.4 to 0.8 are overconfident; only 0.8+ is High

04

Checks

NLI bar holds back good rewrites; number words fail the check

Live demo: from upload to evidence

1 Upload PDF

Choose a contract

2 Ask a question

Route to the right topic

3 Inspect answer + source

Review confidence and cited clause

[TODO] Insert app URL / backup recording

Questions?

Jargonaut makes contract answers simpler — without separating them from their evidence.

[TODO] Add repository / contact details